BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

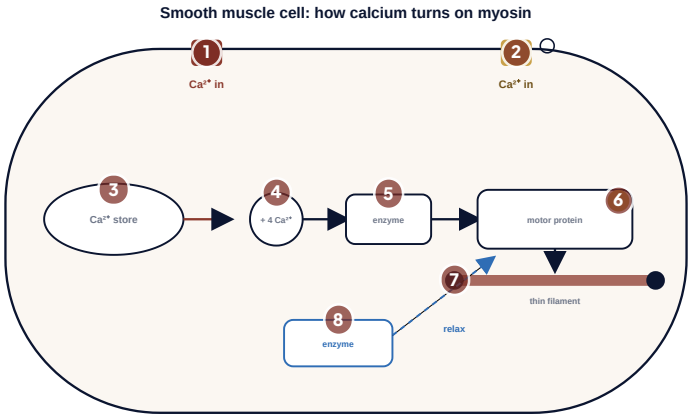
Name _____

A. Smooth muscle: how it contracts and what controls it

YOU SHOULD BE ABLE TO

Smooth muscle contraction mechanism. Trace smooth muscle contraction through calcium entry and release to calmodulin and myosin light chain kinase and explain why the mechanism is slower and more economical than in skeletal muscle.

Smooth muscle types and regulation. Compare single unit and multi unit smooth muscle and list the stimuli that alter smooth muscle tone including stretch and local chemical signals.



Smooth muscle cell: the calcium switch

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____
8 _____

WHAT YOU ARE LOOKING AT

One smooth muscle cell, cut open. The membrane is the outline, the small Ca^{2+} store (SR) sits inside, and the contractile proteins are at the lower right. There is no troponin and no sarcomere, so the switch is different: it is on myosin, and it runs through an enzyme.

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

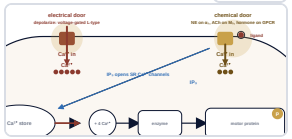
Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

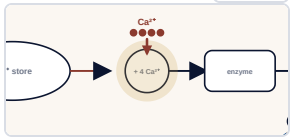
A. Smooth muscle: how it contracts and what controls it (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

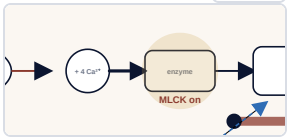
STEP 1



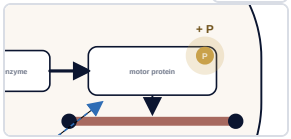
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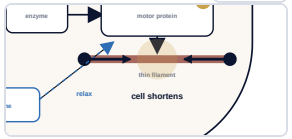
STEP 3



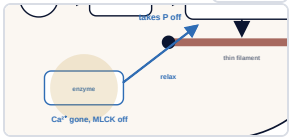
STEP 4



STEP 5

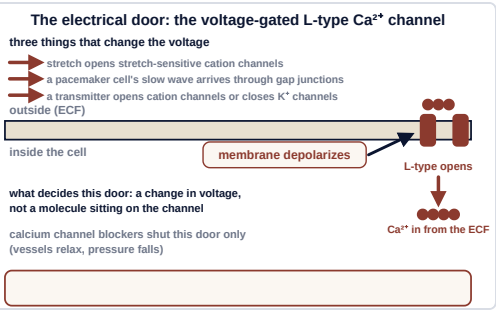


STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

1 Voltage-gated Ca^{2+} channel: how it works



Write the chain in your own words, from what starts it to what it ends in. Then fill the empty box at the bottom of the drawing with the result.

Extra notes and drawings

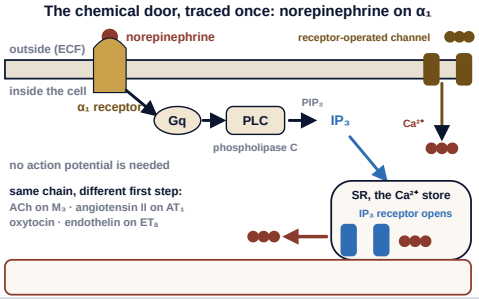
BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

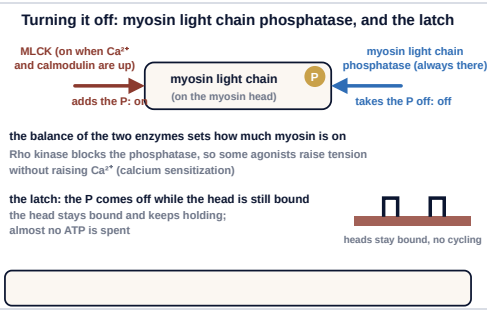
A. Smooth muscle: how it contracts and what controls it (continued)

2 Receptor-operated Ca^{2+} channel: how it works



Write the chain in your own words, from what starts it to what it ends in. Then fill the empty box at the bottom of the drawing with the result.

8 Myosin light chain phosphatase: how it works



Write the chain in your own words, from what starts it to what it ends in. Then fill the empty box at the bottom of the drawing with the result.

WHY SMOOTH MUSCLE IS SLOWER AND CHEAPER TO RUN

Regulation is on

Myosin ATPase is

The latch state

Ca^{2+} mostly comes from

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

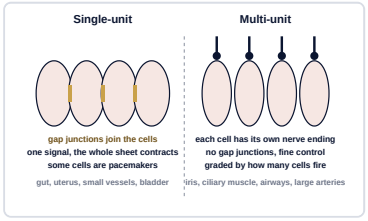
Name _____

A. Smooth muscle: how it contracts and what controls it (continu

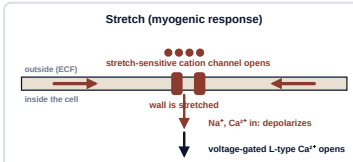
WHAT CHANGES SMOOTH MUSCLE TONE?

Smooth muscle tone

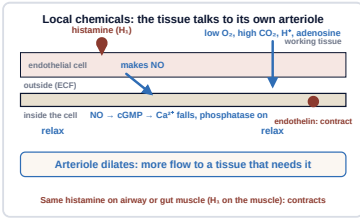
Single-unit vs multi-unit



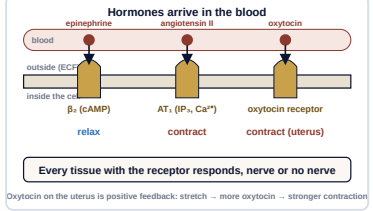
Stretch



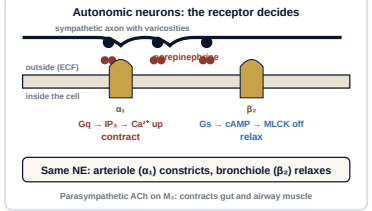
Local chemicals



Hormones



Autonomic neurons



Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

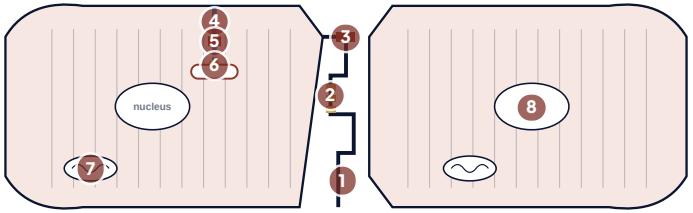
B. Cardiac muscle, and the three muscle types side by side

YOU SHOULD BE ABLE TO

Cardiac muscle properties. Explain how intercalated discs and gap junctions allow the myocardium to act as a functional syncytium and state why cardiac muscle cannot be tetanized.

Comparison of the three muscle types. Build a comparison of skeletal, cardiac, and smooth muscle by calcium source, regulatory protein, control, and speed of contraction.

Two cardiac muscle cells joined at an intercalated disc



1 _____ 5 _____

2 _____ 6 _____

3 _____ 7 _____

4 _____ 8 _____

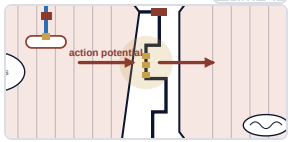
WHAT YOU ARE LOOKING AT

Two cardiac muscle cells end to end. The stepped dark line between them is the intercalated disc, with its gap junction channels and desmosome anchors. Inside the left cell, a T-tubule drops in from the membrane and meets the sarcoplasmic reticulum.

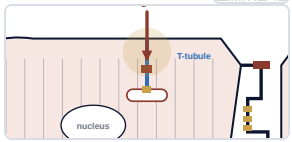
Two cardiac cells and the intercalated disc

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

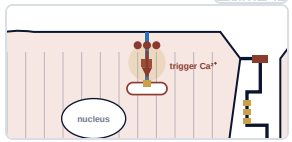
STEP 1



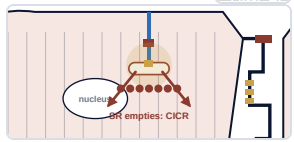
STEP 2



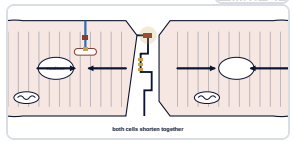
STEP 3



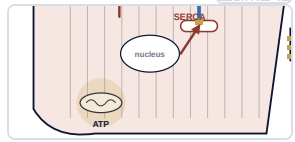
STEP 4



STEP 5



STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

B. Cardiac muscle, and the three muscle types side by side (continued)

COMPARE THE THREE MUSCLE TYPES

	SKELETAL	CARDIAC	SMOOTH
Ca^{2+} source			
Regulatory protein			
Control			
Speed and tetanus			
Cell shape			

C. The adrenal medulla: a sympathetic ganglion that secretes into blood

YOU SHOULD BE ABLE TO

Adrenal medulla. Explain why the adrenal medulla is a modified sympathetic ganglion and compare the duration of its circulating catecholamine effect with direct sympathetic innervation.

FROM THE SPINAL CORD TO EVERY TISSUE

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5

NERVE VERSUS HORMONE

Direct sympathetic nerve: speed and reach

Adrenal medulla: speed and reach

Why both?

BIO 005 Week 9 · The Heart as a Pump

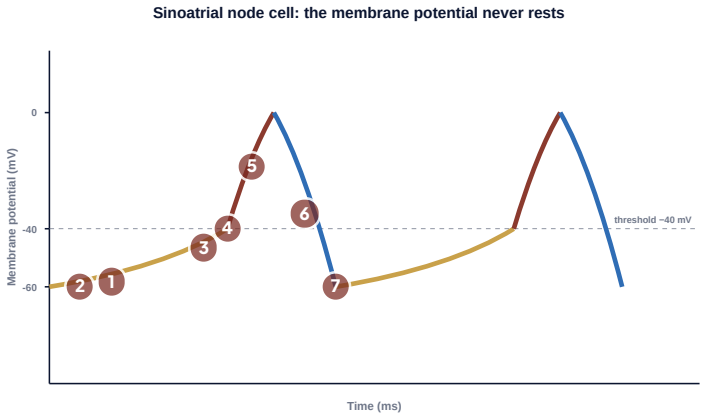
Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

D. The pacemaker and the cardiac action potential

YOU SHOULD BE ABLE TO

Pacemaker potential. Explain the unstable pacemaker potential of the sinoatrial node by the funny current, calcium entry, and potassium exit and state what sets intrinsic heart rate.
Contractile cell action potential. Diagram the ventricular action potential and identify the ion movement responsible for the rapid upstroke, the plateau, and repolarization.
Cardiac refractory period. Explain why the long refractory period of cardiac muscle prevents tetanus and why that matters for pump function.



SA node cell: pacemaker potential

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____

WHAT YOU ARE LOOKING AT

Membrane potential of one SA node cell over two heartbeats. Notice there is no flat resting line: as soon as one beat ends the cell starts drifting toward the next.

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

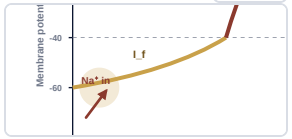
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Name _____

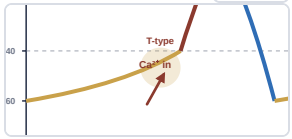
D. The pacemaker and the cardiac action potential (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

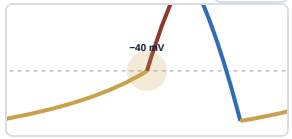
STEP 1



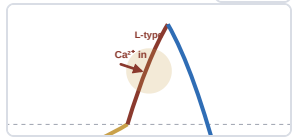
STEP 2



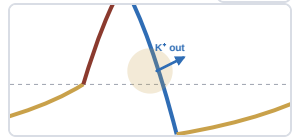
STEP 3



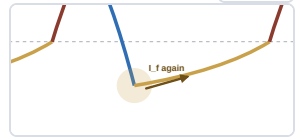
STEP 4



STEP 5

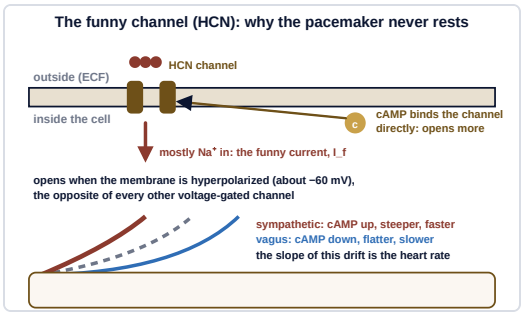


STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

2 Funny current, I_f: how it works

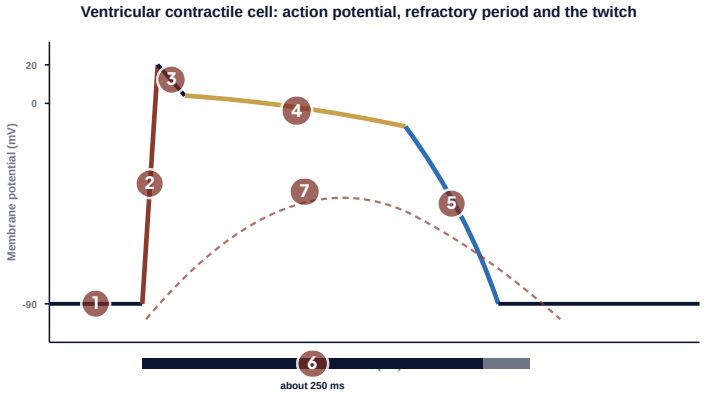


Write the chain in your own words, from what starts it to what it ends in. Then fill the empty box at the bottom of the drawing with the result.

Extra notes and drawings

Name _____

D. The pacemaker and the cardiac action potential (continued)



Ventricular contractile cell: action potential and twitch

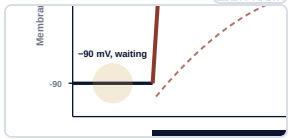
1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____

WHAT YOU ARE LOOKING AT

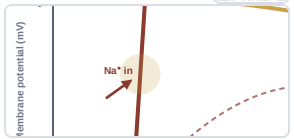
One ventricular contractile cell: membrane potential in solid color, the twitch tension dashed, and the refractory period as the bar underneath. Numbers on the left are millivolts.

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

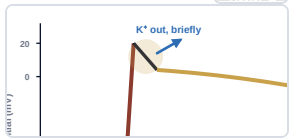
STEP 1



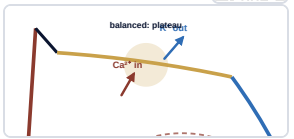
STEP 2



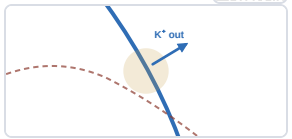
STEP 3



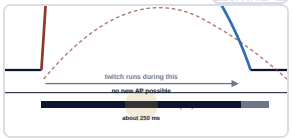
STEP 4



STEP 5



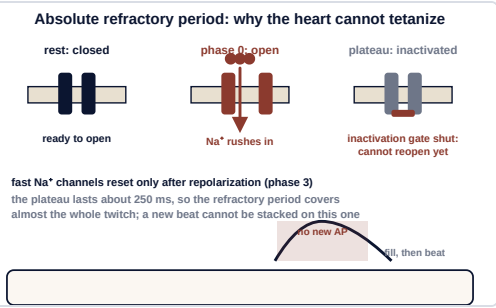
STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

D. The pacemaker and the cardiac action potential (continued)

6 Absolute refractory period: how it works



Write the chain in your own words, from what starts it to what it ends in. Then fill the empty box at the bottom of the drawing with the result.

RATES AND THE REFRACTORY PERIOD

SA node intrinsic rate

AV node backup rate

Purkinje fiber backup rate

Why the heart cannot tetanize

Why that matters

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

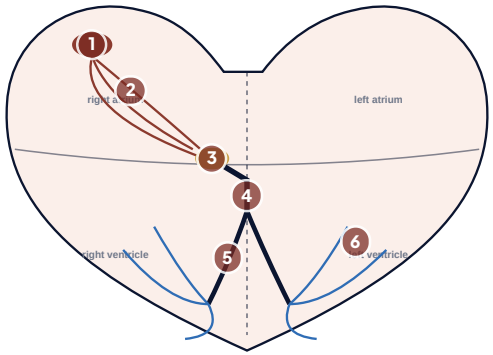
Name _____

E. The wiring and the ECG

YOU SHOULD BE ABLE TO

Conduction system. Trace an impulse through the sinoatrial node, internodal pathways, atrioventricular node, bundle of His, bundle branches, and Purkinje fibers and explain the purpose of the atrioventricular delay.
ECG waves and intervals. Label the P wave, QRS complex, T wave, PR interval, and QT interval and state the electrical event each represents.
ECG interpretation. Calculate heart rate from an ECG strip and identify sinus rhythm, tachycardia, bradycardia, and a first degree or complete heart block.
ECG recording. Record a lead II ECG, measure the intervals, and correlate each wave with the mechanical event that follows it.

The heart's wiring: where the signal starts and where it goes



Conduction system of the heart

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____

WHAT YOU ARE LOOKING AT

The heart from the front, so the patient's right is on your left. Pacemaker and conducting tissue are in color; the dashed line is the septum and the faint curve divides atria from ventricles.

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

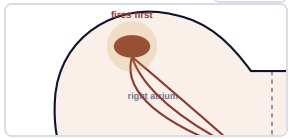
Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

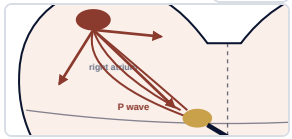
E. The wiring and the ECG (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

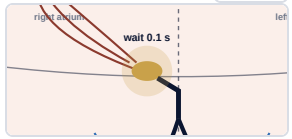
STEP 1



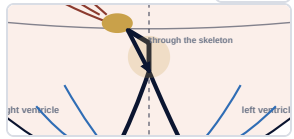
STEP 2



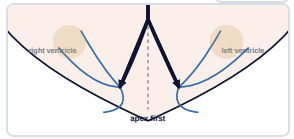
STEP 3



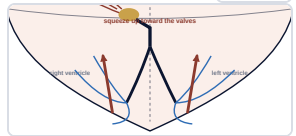
STEP 4



STEP 5



STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

3 Atrioventricular (AV) node: how it works

The AV node delay: about 0.1 s, on purpose

atrial muscle	AV node	bundle of His
big cells, many gap junctions	small cells, few gap junctions	then Purkinje: fastest of all

why it is slow

its upstroke is carried by slow Ca^{2+} channels, not fast Na^{+} channels
small cells and few gap junctions pass current poorly
it is the only electrical path through the fibrous skeleton

what the delay buys

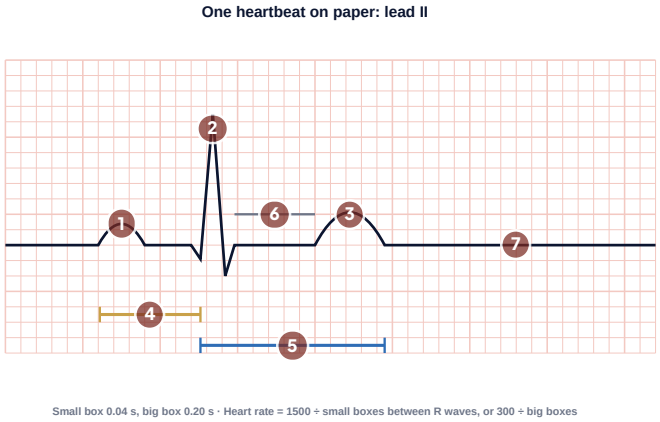
the atria finish contracting and emptying before the ventricles start
on the ECG this is the PR interval; vagus lengthens it, sympathetic shortens it

Write the chain in your own words, from what starts it to what it ends in. Then fill the empty box at the bottom of the drawing with the result.

Extra notes and drawings

Name _____

E. The wiring and the ECG (continued)



Lead II, one beat

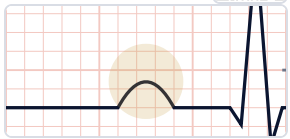
1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____

WHAT YOU ARE LOOKING AT

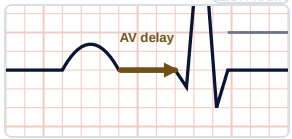
Lead II, one beat, on standard ECG paper. A small box is 0.04 s, a big box 0.20 s. The flat line is the baseline; every wave is read against it.

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

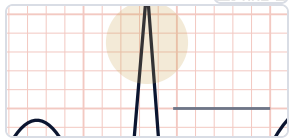
STEP 1



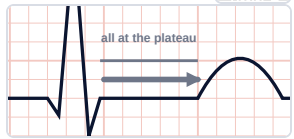
STEP 2



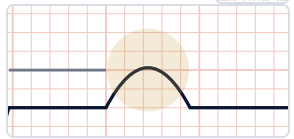
STEP 3



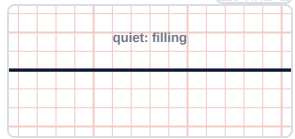
STEP 4



STEP 5



STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

E. The wiring and the ECG (continued)

READ THE STRIP

Heart rate from the strip

Sinus rhythm

Tachycardia

Bradycardia

First-degree heart block

Complete (third-degree) heart block

Wave to mechanical event

F. The cardiac cycle, the loop, and the sounds

YOU SHOULD BE ABLE TO

Cardiac cycle. Order the phases of the cardiac cycle and state the pressure relationship that opens and closes the atrioventricular and semilunar valves in each phase.
Pressure volume loop. Label the four phases of a ventricular pressure volume loop and predict how the loop changes with increased preload, increased afterload, or increased contractility.
Heart sounds. Explain the origin of the first and second heart sounds and relate a murmur to valve stenosis or regurgitation by its timing.
Heart sound and pulse correlation. Auscultate the heart sounds, locate the valve areas, and correlate the sounds with the pulse and with the cardiac cycle phases.

THE CARDIAC CYCLE, PHASE BY PHASE

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5

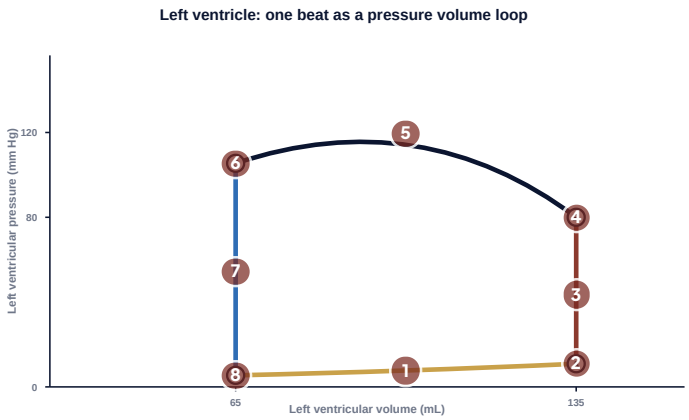
Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

F. The cardiac cycle, the loop, and the sounds (continued)



Left ventricular pressure volume loop

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____
8 _____

WHAT YOU ARE LOOKING AT

Left ventricular pressure (up) against volume (across) for one beat. Time runs counterclockwise around the loop; the corners are the four valve events.

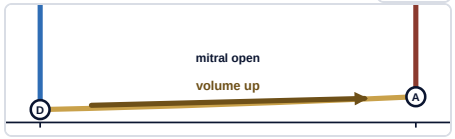
Extra notes and drawings

Name _____

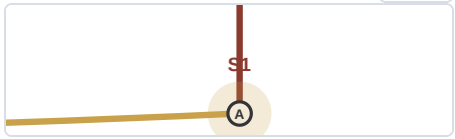
F. The cardiac cycle, the loop, and the sounds (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

STEP 1



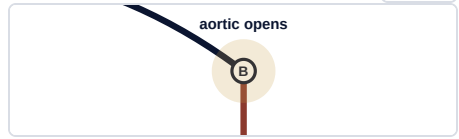
STEP 2



STEP 3



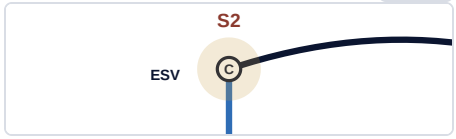
STEP 4



STEP 5



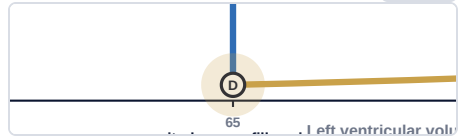
STEP 6



STEP 7



STEP 8



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Extra notes and drawings

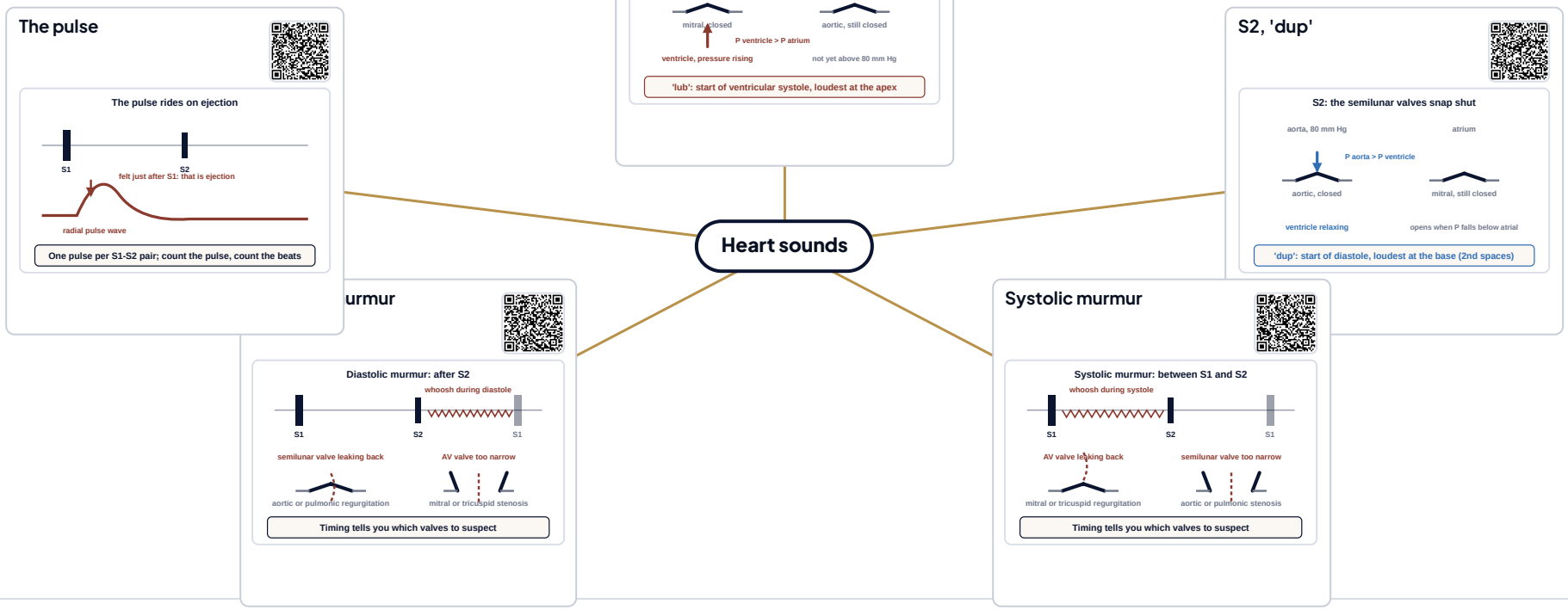
BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

F. The cardiac cycle, the loop, and the sounds (continued)

WHAT YOU HEAR, AND WHEN



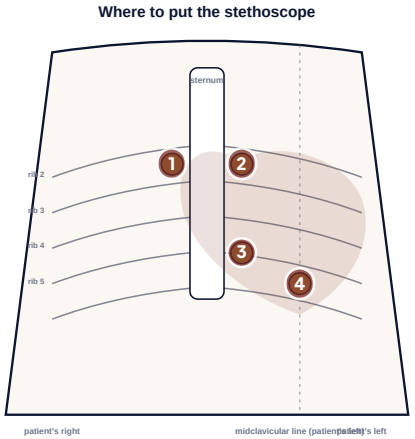
Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

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Name _____

F. The cardiac cycle, the loop, and the sounds (continued)



Valve areas on the chest wall

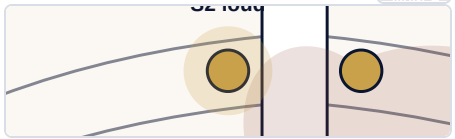
1 _____
2 _____
3 _____
4 _____

WHAT YOU ARE LOOKING AT

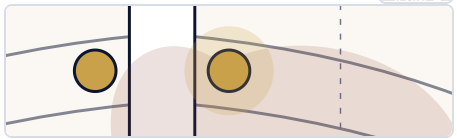
The front of the chest: sternum in the middle, ribs counted from the top, the heart faint behind. Sound travels in the direction the blood flows, so each valve is heard a little downstream of where it sits.

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

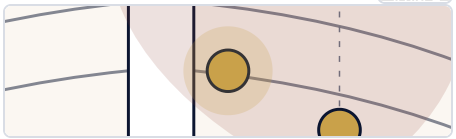
STEP 1



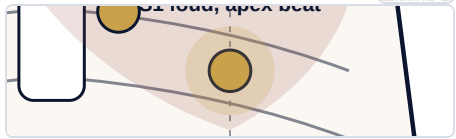
STEP 2



STEP 3



STEP 4



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

G. Stroke volume, cardiac output, and what controls them

YOU SHOULD BE ABLE TO

Stroke volume and ejection fraction. Calculate stroke volume and ejection fraction from end diastolic and end systolic volumes and interpret a reduced ejection fraction.
Cardiac output. Calculate cardiac output and cardiac reserve and predict the effect of a change in heart rate or stroke volume on each.
Frank Starling relationship. State the Frank Starling law and explain how venous return and end diastolic volume determine the force of the next contraction.
Preload afterload and contractility. Distinguish preload, afterload, and contractility and predict how a change in each alters stroke volume.
Autonomic regulation of the heart. Explain how sympathetic and parasympathetic input alter heart rate and contractility at the receptor and second messenger level.

THE NUMBERS (TYPICAL RESTING ADULT)

End-diastolic volume (EDV)

End-systolic volume (ESV)

Stroke volume = EDV − ESV

Ejection fraction = SV ÷ EDV

Cardiac output = HR × SV

Cardiac reserve

Change HR or SV

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

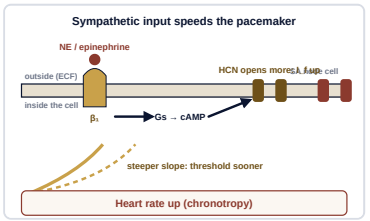
Name _____

G. Stroke volume, cardiac output, and what controls them (cont

WHAT SETS CARDIAC OUTPUT?

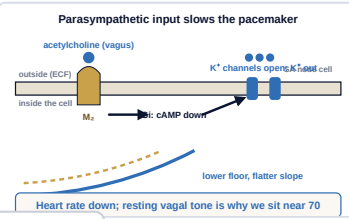
Cardiac output = HR × SV

Heart rate: sympathetic up



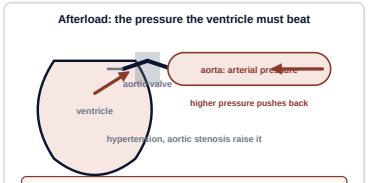
Heart rate up (chronotropy)

Heart rate: parasympathetic down



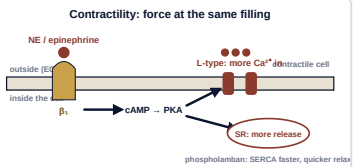
Heart rate down; resting vagal tone is why we sit near 70

Afterload



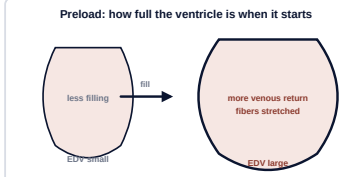
Higher afterload: valve opens later, less ejected, ESV rises

Contractility



Stronger squeeze, ESV falls, stroke volume up (inotropy)

Preload



Frank-Starling: more stretch, more force, bigger stroke volume

SYMPATHETIC INPUT AT THE RECEPTOR LEVEL

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6

Name _____

G. Stroke volume, cardiac output, and what controls them (continued)

FRANK-STARLING IN ONE LINE EACH

The law

Why

What raises venous return

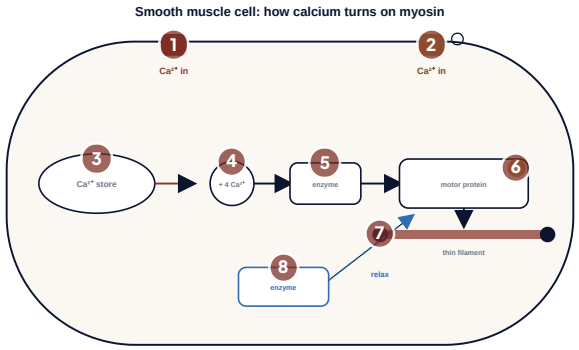
Why it matters

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump STRUCTURE NOTES

A. Smooth muscle: how it contracts and what controls it · Competency Smooth muscle contraction mechanism · Smooth muscle types and regulation · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



Smooth muscle cell: the calcium switch

1 _____

What it is

What triggers it

What it does

Why it matters

2 _____

What it is

What triggers it

What it does

Why it matters

3 _____

What it is

What it does

Why it matters

5 _____

What it is

What it does

Why it matters

6 _____

What it is

What it does

Why it matters

7 _____

What it is

What it does

Why it matters

8 _____

What it is

What it does

Why it matters

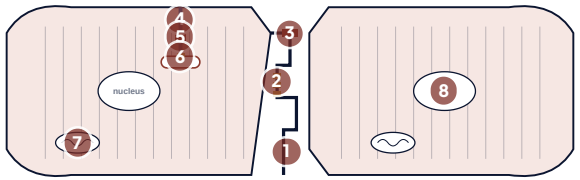
BIO 005 Week 9 · The Heart as a Pump

STRUCTURE NOTES

Cardiac muscle, and the three muscle types side by side · Competency Cardiac muscle properties · Comparison of the three muscle types · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____

Two cardiac muscle cells joined at an intercalated disc



Two cardiac cells and the intercalated disc

1 _____

What it is

What it does

Why it matters

2 _____

What it is

What it does

Why it matters

3 _____

What it is

What it does

Why it matters

6 _____

What it is

What it does

Why it matters

7 _____

What it is

What it does

Why it matters

8 _____

What it is

What it does

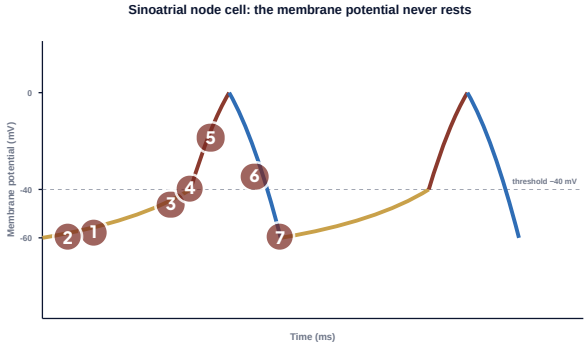
Why it matters

BIO 005 Week 9 · The Heart as a Pump

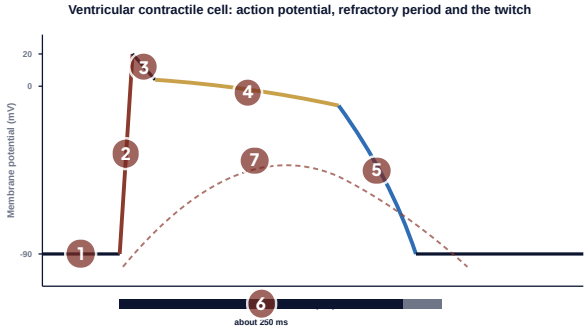
STRUCTURE NOTES

The pacemaker and the cardiac action potential · Competency Pacemaker potential · Contractile cell action potential · Cardiac refractory period · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



SA node cell: pacemaker potential



Ventricular contractile cell: action potential and twitch

1 _____
What it is

What it does

Why it matters

2 _____
What it is

What it does

Why it matters

3 _____
What it is

What it does

Why it matters

3 _____
What it is

What it does

Why it matters

4 _____
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What it does

Why it matters

4 _____
What it is

What it does

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What it does

Why it matters

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Why it matters

7 _____
What it is

What it does

Why it matters

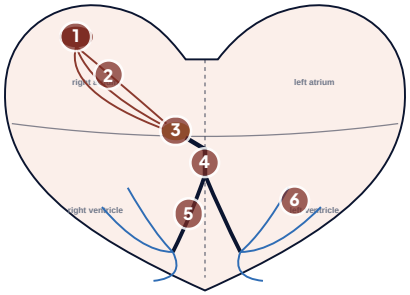
BIO 005 Week 9 · The Heart as a Pump

STRUCTURE NOTES

The wiring and the ECG · Competency Conduction system · ECG waves and intervals · ECG interpretation · ECG recording · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

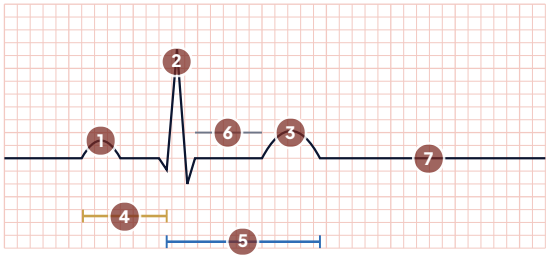
Name _____

The heart's wiring: where the signal starts and where it goes



Conduction system of the heart

One heartbeat on paper: lead II



Small box 0.04 s, big box 0.20 s · Heart rate = 1500 ÷ small boxes between R waves, or 300 ÷ big boxes

Lead II, one beat

1 _____
What it is
What it does
Why it matters

2 _____
What it is
What it does
Why it matters

3 _____
What it is
What it does
Why it matters

4 _____
What it is
What it does
Why it matters

5 _____
What it is
What it does
Why it matters

6 _____
What it is
What it does
Why it matters

1 _____
What it is
What it does
Why it matters

4 _____
What it is
What it does
Why it matters

5 _____
What it is
What it does
Why it matters

6 _____
What it is
What it does
Why it matters

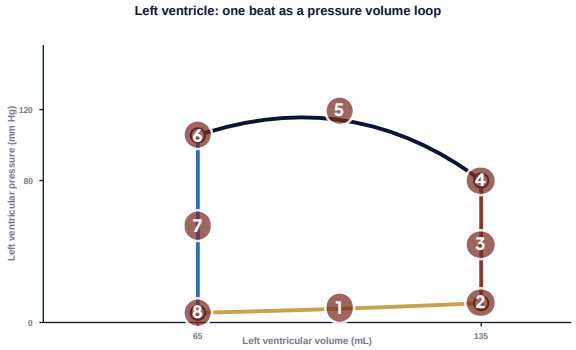
7 _____
What it is
What it does
Why it matters

BIO 005 Week 9 · The Heart as a Pump

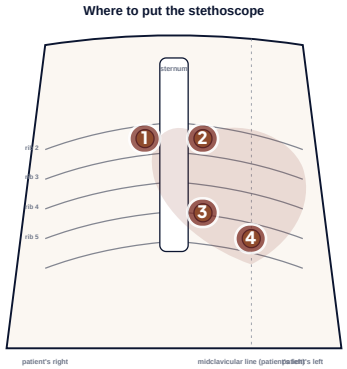
STRUCTURE NOTES

F. The cardiac cycle, the loop, and the sounds · Competency Cardiac cycle · Pressure volume loop · Heart sounds · Heart sound and pulse correlation · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



Left ventricular pressure volume loop



Valve areas on the chest wall

1 _____
What it is
What it does
Why it matters

2 _____
What it is
What it does
Why it matters

3 _____
What it is
What it does
Why it matters

4 _____
What it is
What it does
Why it matters

5 _____
What it is
What it does
Why it matters

6 _____
What it is
What it does
Why it matters

7 _____
What it is
What it does
Why it matters

2 _____
What it is
What it does
Why it matters

3 _____
What it is
What it does
Why it matters

4 _____
What it is
What it does
Why it matters

2 _____

What it is

What it does

Why it matters

8 _____

What it is

What it does

Why it matters

1 _____

What it is

What it does

Why it matters